

## Supplementary Material

**Supplementary Table S1.** Demographic and clinical data of STS patients and controls samples included in the study.

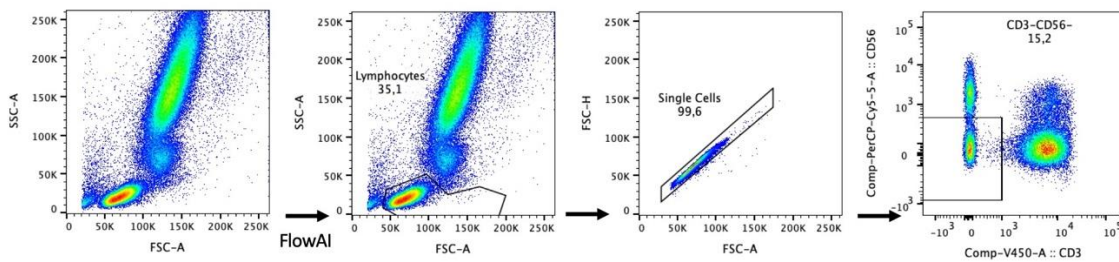
| Characteristic, unit                                              | N (%) or mean (SD) |            |
|-------------------------------------------------------------------|--------------------|------------|
|                                                                   | Ctrl               | STS        |
| Samples                                                           | 25                 | 29 (100%)  |
| Age, years                                                        | 49 ± 10            | 54 ± 14    |
| Sex (% of females)                                                | 15 (60%)           | 16 (55%)   |
| Disease status                                                    |                    |            |
| Non-metastatic                                                    |                    | 14 (48.2%) |
| Primary                                                           |                    | 13 (45.8%) |
| Recurrence                                                        |                    | 1 (3.4%)   |
| Metastatic                                                        |                    | 15 (51.7%) |
| Primary                                                           |                    | 13 (45.8%) |
| Recurrence                                                        |                    | 2 (6.9%)   |
| Primary anatomical localization                                   |                    |            |
| Extremity (Lower limb)                                            |                    | 14 (48.2%) |
| Trunk (not retroperitoneal)                                       |                    | 8 (36.4%)  |
| Thorax                                                            |                    | 4 (13.8%)  |
| Pelvis                                                            |                    | 2 (6.9%)   |
| Liver                                                             |                    | 1 (3.4%)   |
| Trunk unspecified                                                 |                    | 1 (3.4%)   |
| Retroperitoneal                                                   |                    | 3 (10.3%)  |
| Gynecological region                                              |                    | 4 (13.8%)  |
| Uterus                                                            |                    | 3 (10.3%)  |
| Spermatic cord                                                    |                    | 1 (3.4%)   |
| Lineage of cell differentiation                                   |                    |            |
| Leiomyosarcoma                                                    |                    | 6 (20.7%)  |
| Liposarcoma                                                       |                    | 4 (13.8%)  |
| Undifferentiated sarcoma                                          |                    | 7 (24.1%)  |
| Synovial sarcoma                                                  |                    | 6 (20.7%)  |
| Other                                                             |                    | 6 (20.7%)  |
| Malignant peripheral nerve sheath tumor                           |                    | 2 (6.9%)   |
| Hemangiosarcoma                                                   |                    | 1 (3.4%)   |
| Clear cell sarcoma                                                |                    | 1 (3.4%)   |
| Embryonal sarcoma                                                 |                    | 1 (3.4%)   |
| Endometrial stromal sarcoma                                       |                    | 1 (3.4%)   |
| Therapy                                                           |                    |            |
| Anthracycline-based therapy                                       |                    | 7 (16.4%)  |
| Anthracycline-based therapy followed by trabectedin-based therapy |                    | 9 (34.5%)  |
| Trabectedin-based therapy                                         |                    | 3 (14.5%)  |
| Other                                                             |                    | 3 (20%)    |
| Not applicable                                                    |                    | 7 (14.5%)  |
| Peripheral Immune Profile                                         |                    |            |
| P1 – “Immune high”                                                |                    | 9 (56.4%)  |
| P2 – “Immune intermediate”                                        |                    | 8 (36.4%)  |
| P3 – “Immune low”                                                 |                    | 12 (7.3%)  |

**Supplementary Table S2.** Detailed description of the fluorochrome-conjugated monoclonal antibodies used to perform multiparametric flow cytometry analysis.

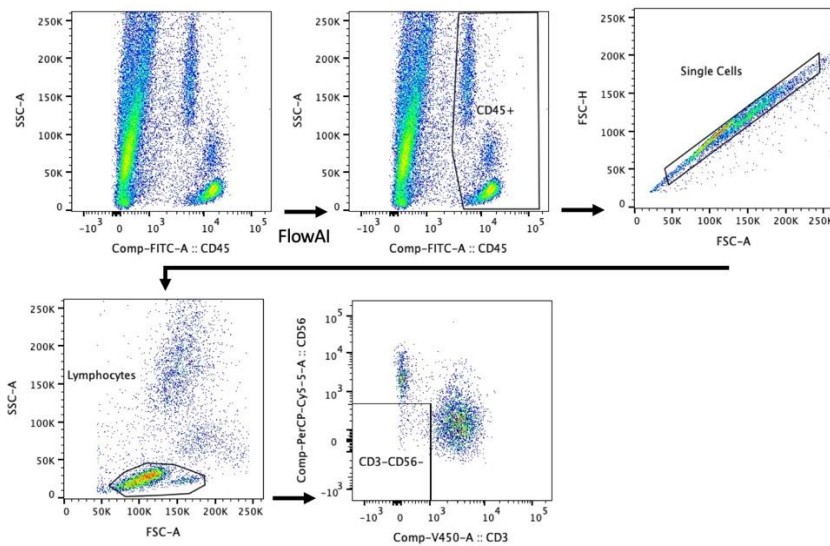
| Antibody            | Conjugate   | Clone   | Brand          | Cat#   | RRDI        |
|---------------------|-------------|---------|----------------|--------|-------------|
| CD3                 | V450        | ICHT1   | BD Horizon™    | 561416 | AB_10612021 |
| CD4                 | V500        | RPA-T4  | BD Horizon™    | 560768 | AB_1937323  |
| CD8                 | APC-H7      | SK1     | BD™            | 641400 | AB_1645736  |
| CD45                | FITC        | HI30    | BD Pharmingen™ | 555482 | AB_395874   |
| CD56                | PerCP-Cy5.5 | B159    | BD Pharmingen™ | 560842 | AB_2033964  |
| CD161               | PE-Cy7      | HP-3G10 | Biolegend™     | 339918 | AB_11126745 |
| TCR $\gamma/\delta$ | PE          | B1      | Biolegend™     | 331210 | AB_1089218  |

**Supplementary File S3.** Representative images of the manual gating strategy used to identify lymphocyte populations expressing CD3 and/or CD56 in peripheral blood (**A**) and tumor (**B**) samples using FlowJo® v.10.9 software. First, FlowAI plugin was used to remove low-quality events. **A**) lymphocytes were gated based on FSC-A vs. SSC-A, doublets were excluded, and CD3<sup>-</sup>CD56<sup>-</sup> events were used with a “make not-gate” strategy to isolate CD3<sup>+</sup> and/or CD56<sup>+</sup> populations. **B**) CD45<sup>+</sup> events were selected, doublets excluded, lymphocytes gated by FSC-A vs. SSC-A, and the same “make not-gate” approach was applied to CD3<sup>-</sup>CD56<sup>-</sup> events to identify CD3<sup>+</sup> and/or CD56<sup>+</sup> cells.

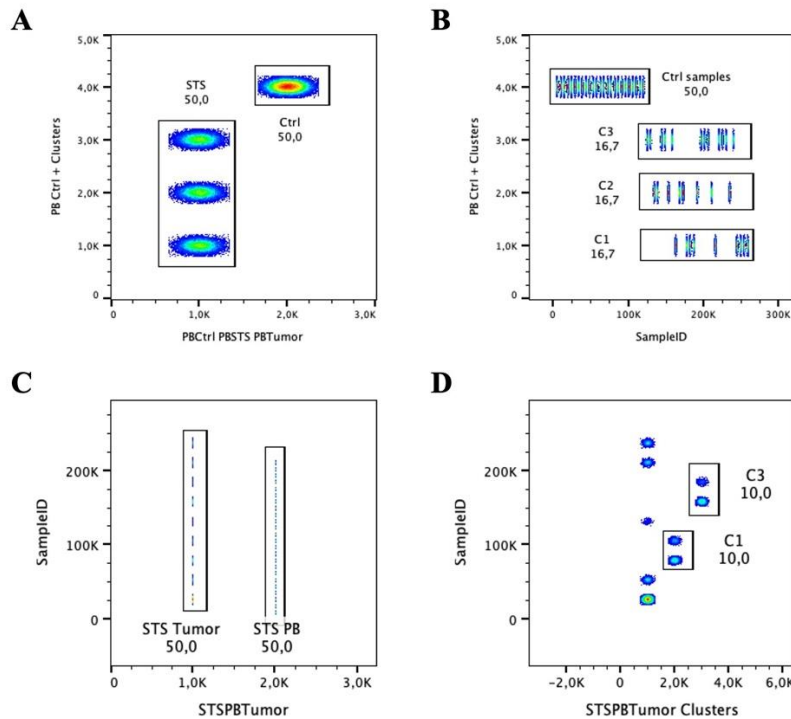
**A**



**B**



**Supplementary File S4.** Representative images demonstrating the selection of experimental groups in the concatenated files for peripheral blood (**A, B**) and tumor (**C, D**) samples using FlowJo® v10.9 software.



**Supplementary File S5.** Survival analysis of STS peripheral immunotypes. Kaplan–Meier curves were generated for STS patients categorized according to previous established peripheral immune profiles, P1 (“immune high”), P2 (“immune intermediate”), and P3 (“immune low”) immunotypes. Survival analysis was assessed using time-to-event from the date of sample collection to either the date of death or the end of follow-up, referred to as TAC (months). Censored events were identified as a cross in the respective curves. Log-rank test was used to compare high and low curves, with significance set at  $p < 0.05$ . *Legend: TAC – time after collection.*

